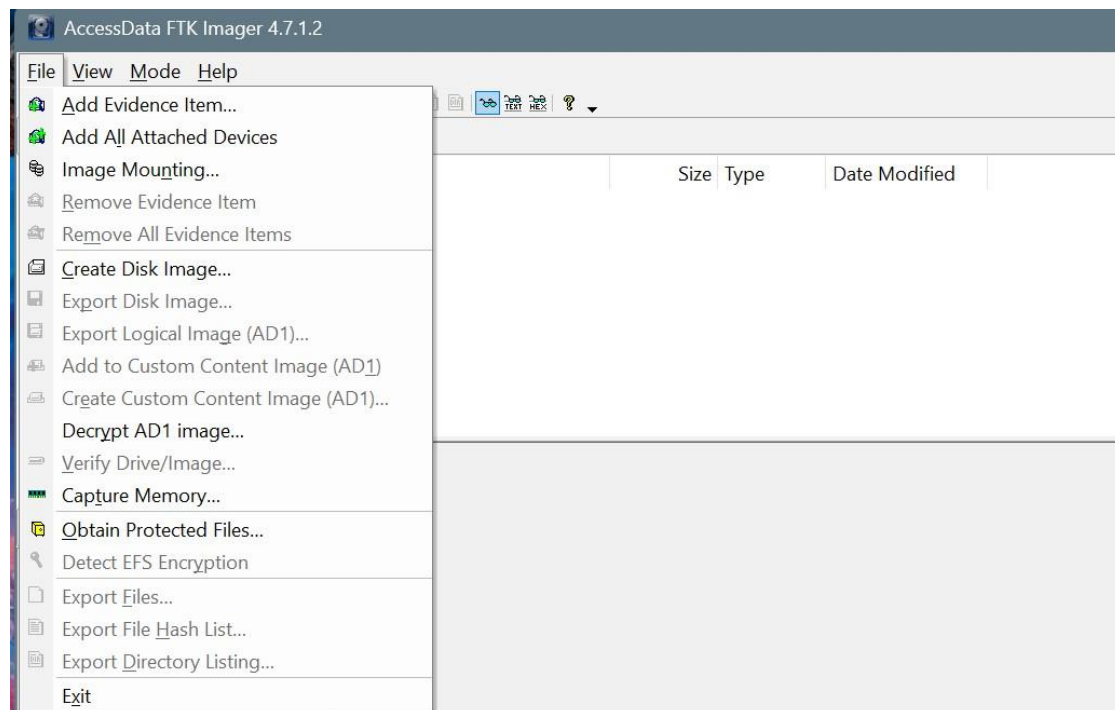




FTK Imager File menu showing the Capture Memory option.

## Ex. No 1: Evidence Acquisition Using AccessData FTK Imager

Date:

### Description

Forensic Toolkit or FTK is a computer forensics software product made by AccessData. This is a Windows based commercial product. For forensic investigations, the same development team has created a free version of the commercial product with fewer functionalities. This FTK Imager tool is capable of both acquiring and analyzing computer forensic evidence.

The evidence FTK Imager can acquire can be split into two main parts. They are:

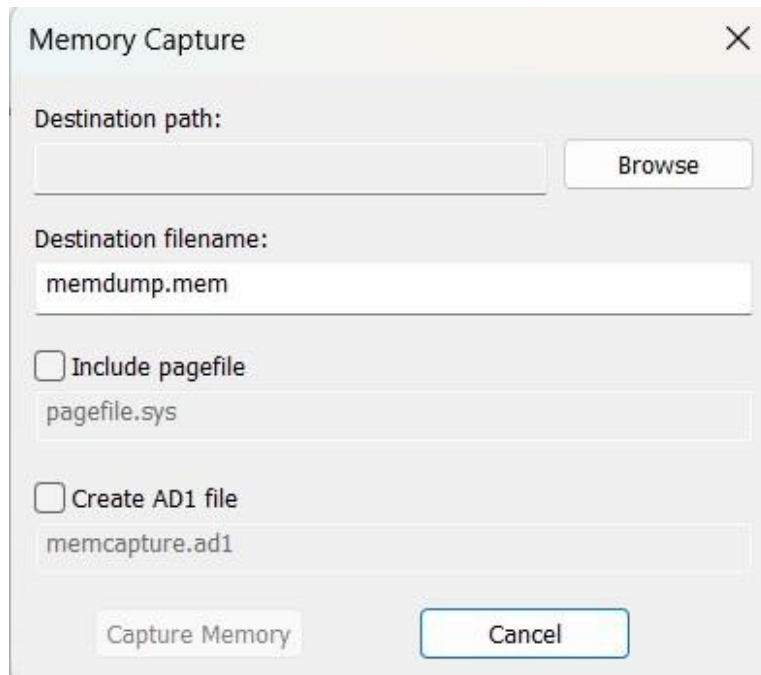
- Acquiring volatile memory
- Acquiring non-volatile memory (Hard disk)

There are two possible ways this tool can be used in forensics image acquisitions:

- Using FTK Imager portable version in a USB pen drive or HDD and opening it directly from the evidencemachine. This option is most frequently used in live data acquisition where the evidence PC/laptop is switched on.
- Installing FTK Imager on the investigator's laptop. In this case the source disk should be mounted into theinvestigator's laptop via write blocker. The write blocker prevents data being modified in the evidence source disk while providing read-only access to the investigator's laptop. This helps to maintain the integrity of the source disk.

### Acquiring volatile memory using FTK Imager

The FTK Imager tool helps investigators to collect the complete volatile memory (RAM) of a computer. The following steps show how to do this. Open FTK Imager and navigate to the volatile memory icon (Capture Memory).



Memory Capture window.

## Acquiring Volatile Memory

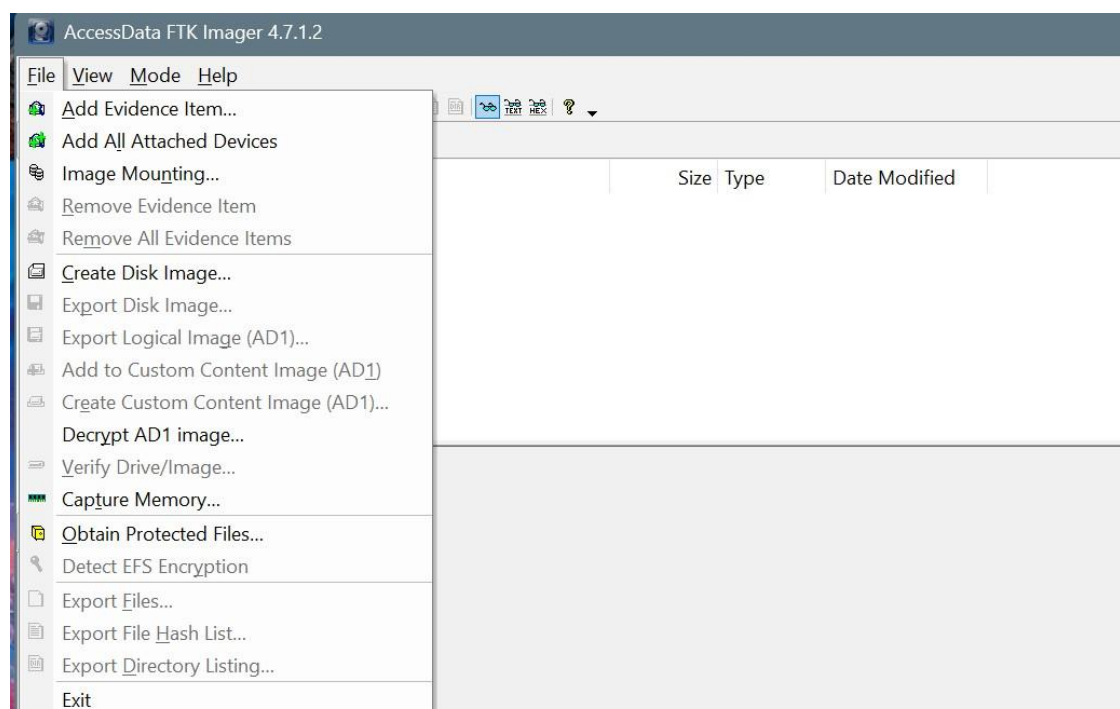
Navigate to the destination location where you need to save the captured volatile memory and create a file name.

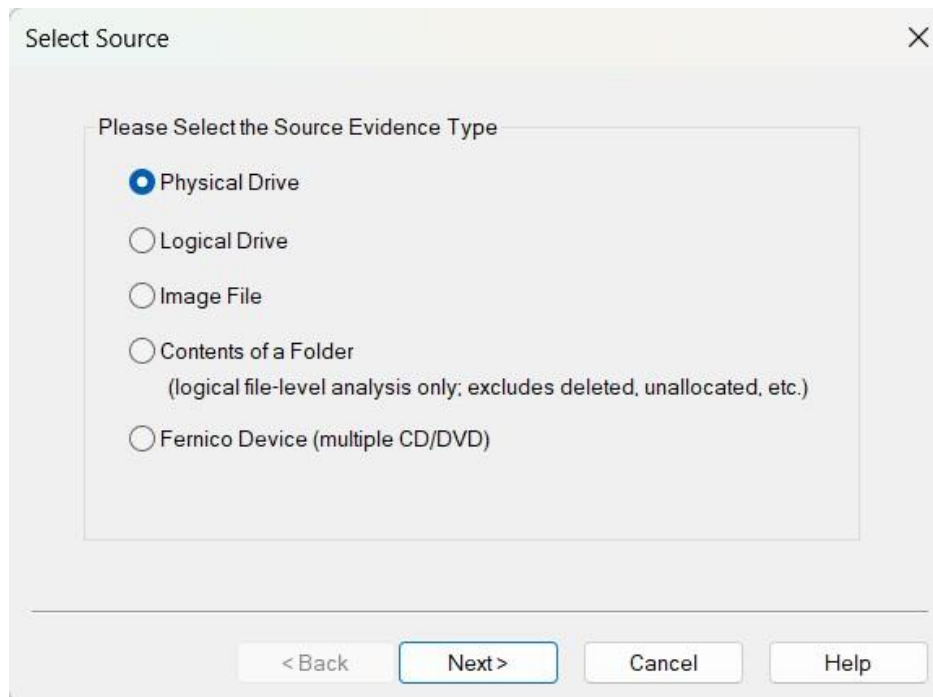
**NOTE:** This tool provides options to include pagefile and AD1 files when acquiring the volatile memory.

**Pagefile:** The pagefile (pagefile.sys) is used in Windows operating systems as volatile memory due to limitation of physical random-access memory (RAM). It is located under the “C” partition ready to use as volatile memory when the existing RAM capacity is exceeded. Therefore, it is recommended to capture and collect this file in the acquisition.

**AD1 file:** AD1 is the FTK Imager image file. The investigator has the option to create an AD1 file for later use.

Clicking the “Capture Memory” button will start acquiring the volatile memory.





Select Source – Physical Drive selected.

## Acquiring non-volatile memory (Disk Image) using FTK Imager

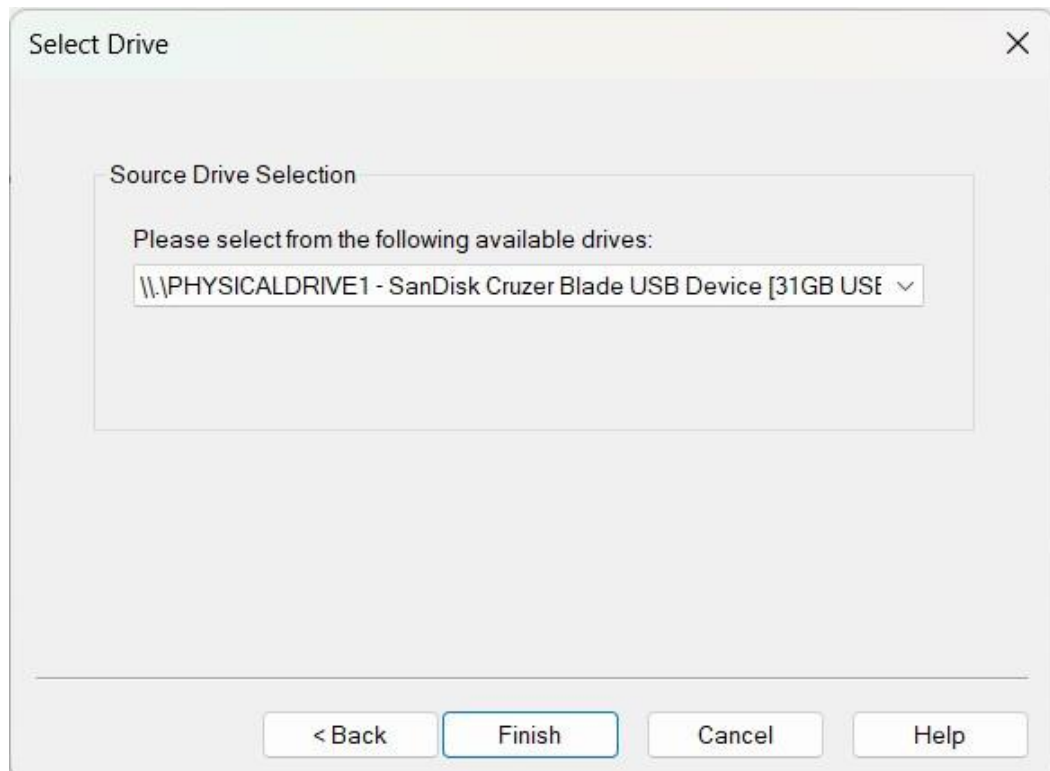
As previously stated, this same tool can be used to collect a disk image as well. Open FTK Imager and navigate to “Create Disk Image”.

Now select the source that you need to acquire.

**NOTE:** FTK Imager is capable of acquiring physical drives (physical hard drives), logical drives (partitions), image files, contents of a folder, or CDs/DVDs. Investigators can connect external HDDs into the collection computer via write blocker and use the “logical drive” option to select the mounted HDD as a partition.

### Collecting Physical Drives

Select the “Physical Drive” option. Select the drive you need to acquire and click “Finish”.



Selected SanDisk Cruzer Blade USB device.

## Selecting the Physical Drive

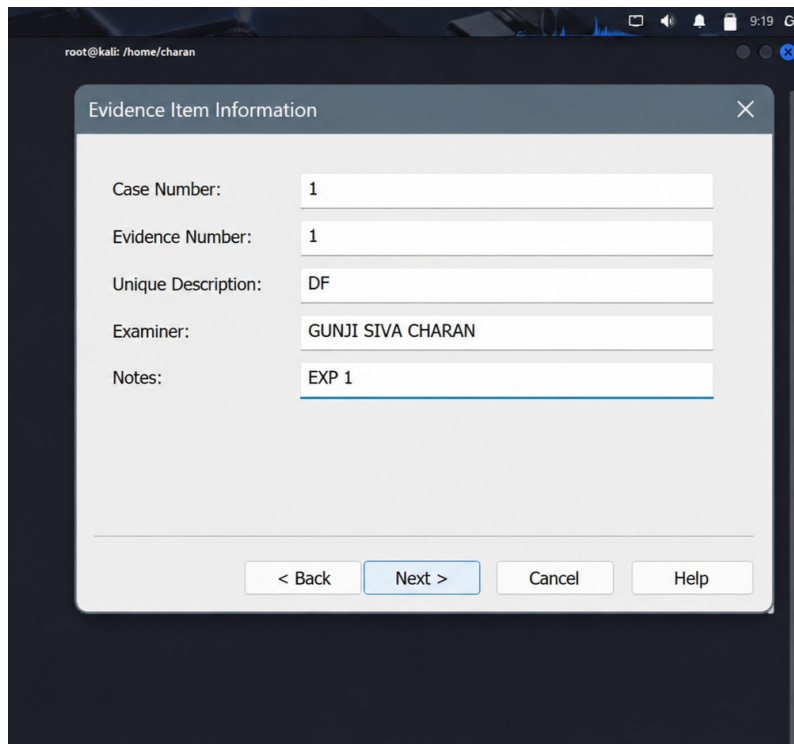
**Raw (dd):** This is the image format most commonly used by modern analysis tools. These raw file formatted images do not contain headers, metadata, or magic values. The raw format typically includes padding for memory ranges that were intentionally skipped or could not be read by the acquisition tool, helping maintain spatial integrity.

**SMART:** This file format is designed for Linux file systems. This format keeps the disk images as pure bitstreams with optional compression.

**E01:** This format is a proprietary format developed by Guidance Software's EnCase. This format compresses the image file and includes case information and hash information.

**AFF:** Advance Forensic Format (AFF) was developed to provide a disk image format that does not lock the user into a proprietary format that may prevent proper analysis.

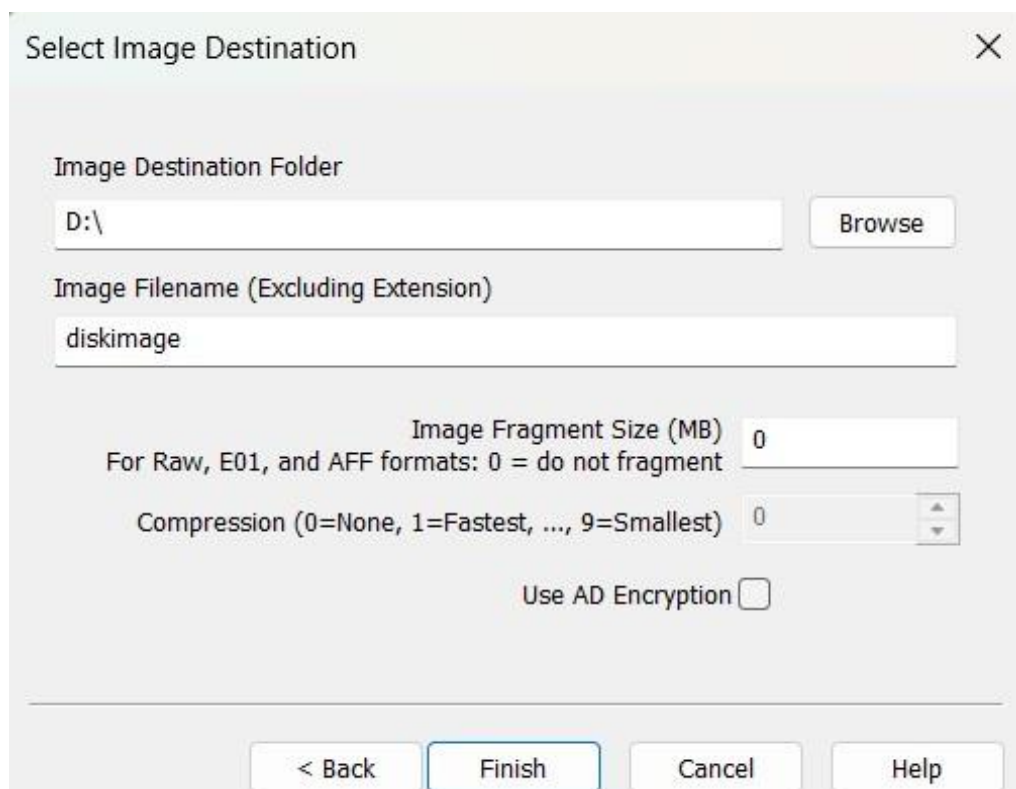
Now enter the case details.



A screenshot of a terminal window with a dark background. A terminal prompt shows 'root@kali: /home/charan'. Overlaid on the terminal is a light gray dialog box titled 'Evidence Item Information' with a close button (X) in the top right corner. The dialog contains five input fields: 'Case Number:' with '1', 'Evidence Number:' with '1', 'Unique Description:' with 'DF', 'Examiner:' with 'GUNJI SIVA CHARAN', and 'Notes:' with 'EXP 1'. At the bottom of the dialog are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'.

| Field               | Value             |
|---------------------|-------------------|
| Case Number:        | 1                 |
| Evidence Number:    | 1                 |
| Unique Description: | DF                |
| Examiner:           | GUNJI SIVA CHARAN |
| Notes:              | EXP 1             |

Case details entered: Case Number 1, Evidence Number 1, Description df, Examiner Rakesh, Notes exp 1.



A screenshot of a 'Select Image Destination' dialog box. It has a title bar with a close button (X). The dialog contains several fields and controls: 'Image Destination Folder' with a text box containing 'D:\' and a 'Browse' button; 'Image Filename (Excluding Extension)' with a text box containing 'diskimage'; 'Image Fragment Size (MB)' with a text box containing '0' and a note 'For Raw, E01, and AFF formats: 0 = do not fragment'; 'Compression (0=None, 1=Fastest, ..., 9=Smallest)' with a spinner box set to '0'; and a checkbox for 'Use AD Encryption' which is unchecked. At the bottom are four buttons: '< Back', 'Finish', 'Cancel', and 'Help'.

| Field                                            | Value                    |
|--------------------------------------------------|--------------------------|
| Image Destination Folder                         | D:\                      |
| Image Filename (Excluding Extension)             | diskimage                |
| Image Fragment Size (MB)                         | 0                        |
| Compression (0=None, 1=Fastest, ..., 9=Smallest) | 0                        |
| Use AD Encryption                                | <input type="checkbox"/> |

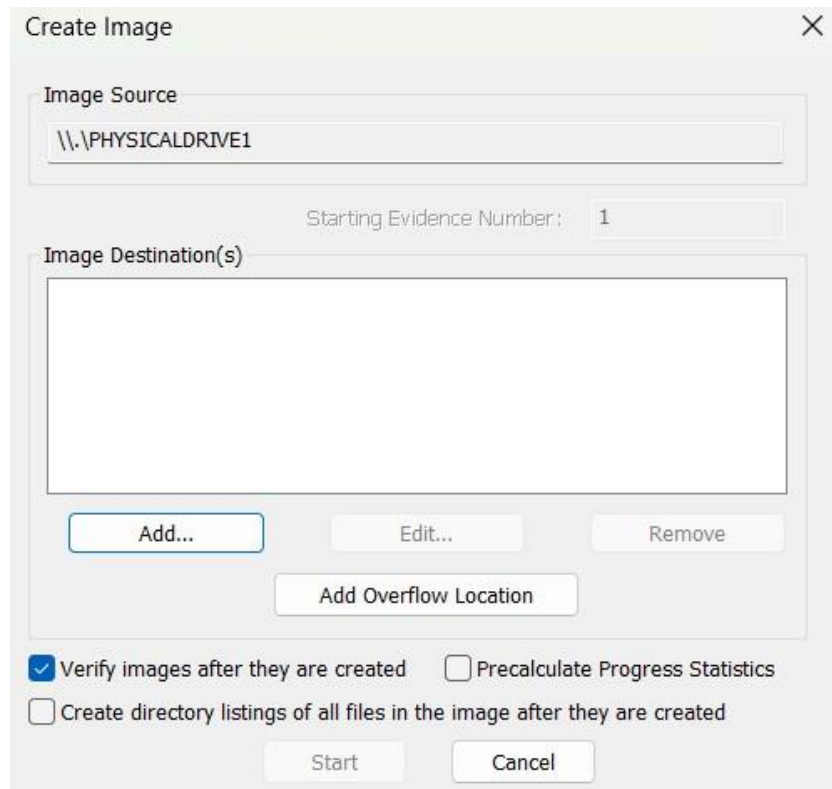
Image destination set to D:\ and filename set to diskimage.

## Entering Evidence Item Information

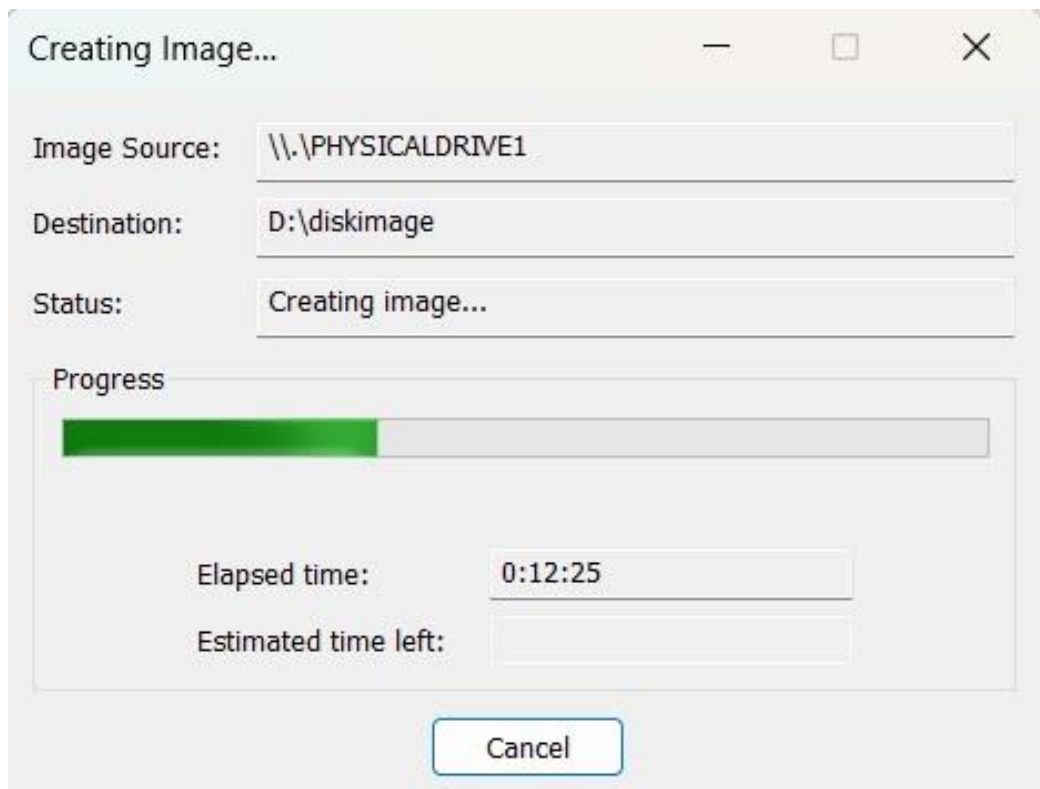
Add an image destination (where the image file will be saved), image file name and fragment size.

**Image Fragment Size (MB):** This option will separate the image file into multiple images and save them in the same destination. If only one file is needed instead of multiple fragmented images, set the image fragment size to “0”.

Select the “verify images after they are created” option. This verifies the hash values once the image has been created. In order to ensure integrity, it is recommended to use this option.



Create Image window with Verify images after they are created selected.



Disk image acquisition in progress.

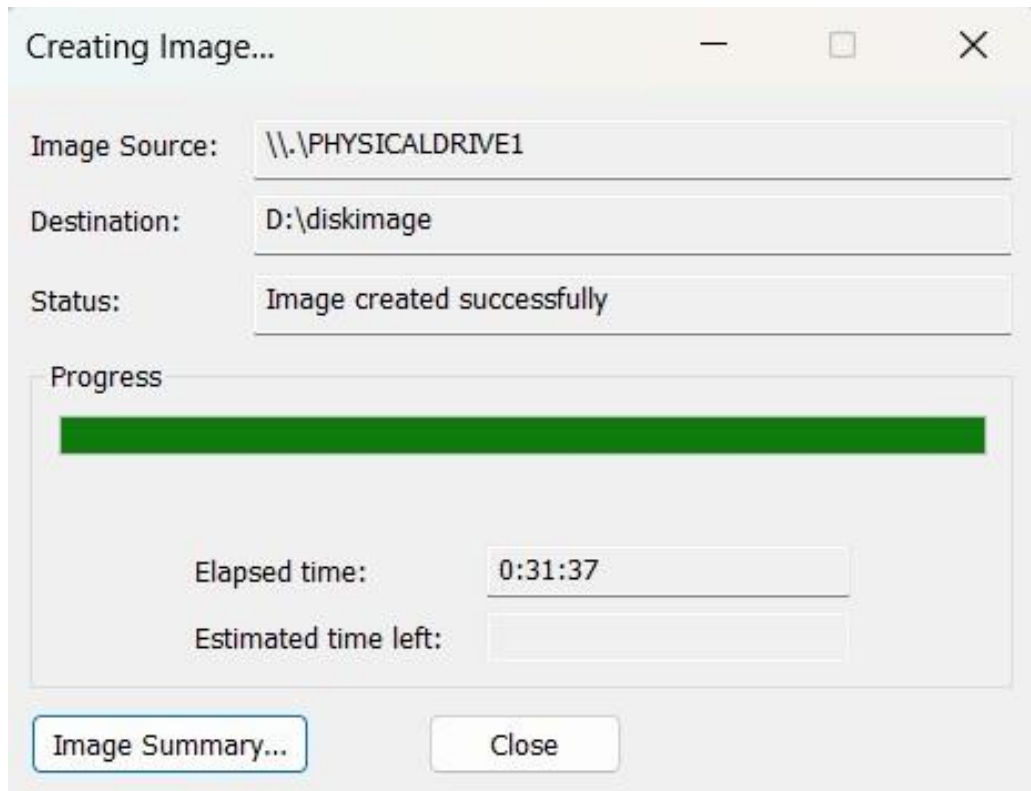


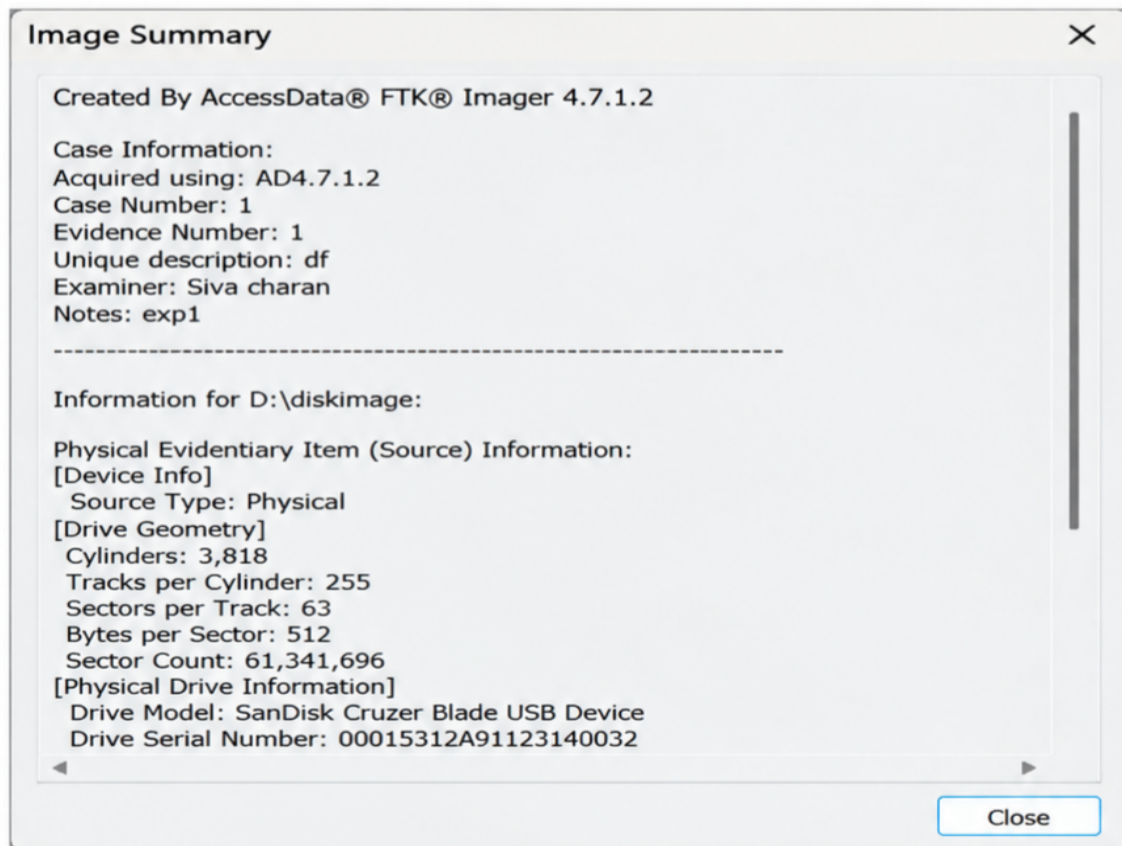
Image created successfully.

## Creating the Disk Image

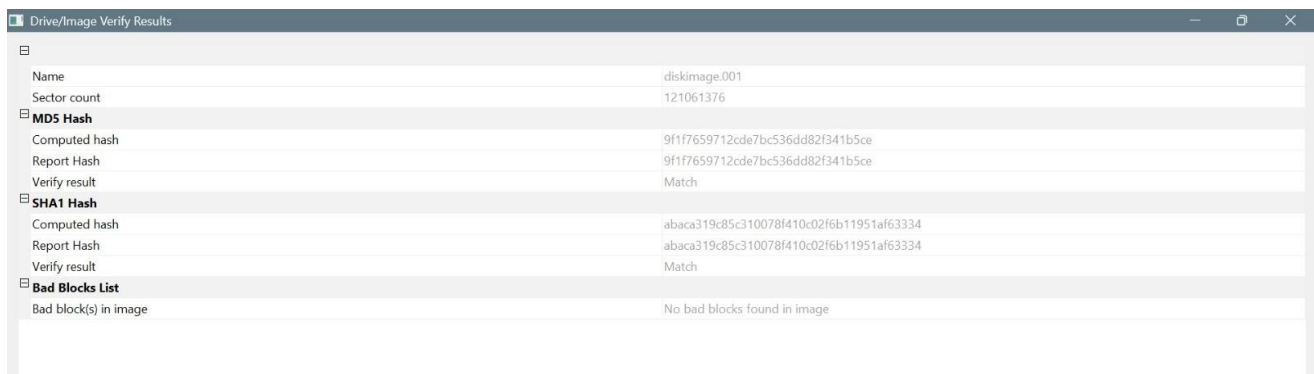
Click "Start" to start acquiring.

Once acquiring is complete, FTK Imager creates the image and associated acquisition information.

Image Summary showing the acquisition and source information.



Drive/Image Verify Results – MD5 and SHA1 Match; no bad blocks found.



## Image Summary and Verification Results

The Image Summary records the evidence acquisition details and physical drive information.

After verification, the MD5 and SHA1 computed hashes match the report hashes. No bad blocks were found in the image.

**Result:** The physical drive image was successfully acquired using AccessData FTK Imager, and the verification results show matching hash values, confirming the integrity of the acquired image.